

Ideas from the East

Medieval astronomers in India developed highly sophisticated mathematical tools for making astronomical calculations. Around 525 CE, the Indian mathematician Aryabhata put forward the idea that Earth rotates on its axis, correctly explaining the apparent movement of the stars. Much of this knowledge passed to astronomers in the Islamic world, who improved on existing theories and refined calculations of how the planets moved within spheres. They also perfected the use of devices called astrolabes, which allowed them to measure the positions of the Sun and stars.



Brass armillary sphere, made in Italy, 1554

The Ptolemaic system

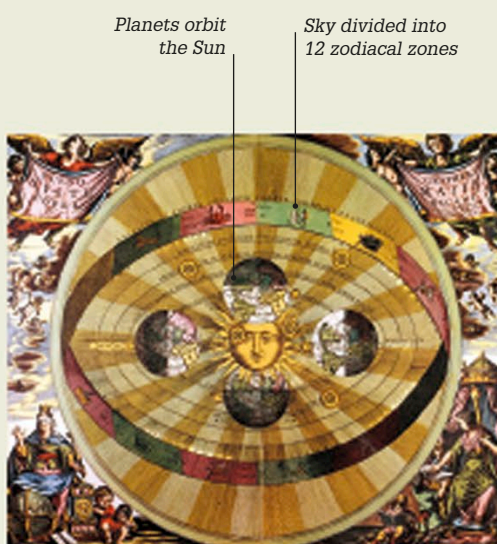
Early Greek astronomers explained the movements of planets by suggesting they orbited within concentric spheres around Earth. This theory was worked out in detail by Ptolemy of Alexandria (c 100–170 CE). To explain oddities in planetary motion, he used a system of epicycles (small circles) within which planets revolved, while at the same time orbiting in larger spheres around Earth. Complicated models, called armillary spheres, were made to illustrate Ptolemy's system.

Refracting telescope

In 1609, the Italian astronomer Galileo Galilei built a telescope. He was not its inventor, but he was the first person to use a telescope for astronomical purposes. Its greatly increased magnification meant Galileo was able to discover four new satellites of Jupiter and to study sunspots for the first time.



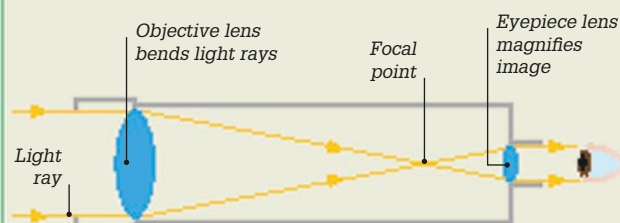
Galileo with his telescope, c 1620



The Copernican Universe, Dutch engraving, 17th century

The Copernican Universe

Polish astronomer Nicolaus Copernicus (1473–1543) disagreed with Ptolemy's views on the Universe. He devised a model, known as a heliocentric system, in which Earth and the planets moved in orbits, with the Sun (rather than Earth) at the center of the solar system.



How it works

Galileo used a refracting telescope, with lenses to gather light and to produce a magnified image. Most modern refracting telescopes are designed as shown above (and work slightly differently from Galileo's). These have an objective lens at one end that gathers light from far objects and refracts (bends) it to a focal point, producing an image. The light then passes through an eyepiece that magnifies the image.

c 130 BCE

Hipparchus, a Greek from Nicaea (now Iznik, Turkey), devised the first accurate star map. He used Ptolemy's model to predict lunar and solar eclipses.



Moon

c 975

Llobet of Barcelona wrote a work introducing the astrolabe to Europe. This device, well known in the Islamic world, calculated the position of the Sun and stars.

1259

An astronomical observatory was built at Maragha, Iran. It allowed Islamic astronomers to make highly accurate measurements of the planets and stars from which to compile charts and tables.

1543

Copernicus published *On the Revolutions of the Celestial Spheres*, setting out his model of a solar system with Earth orbiting the Sun.